

Figure 3.3. Einstein and Bohr, c. 1930.

At the next Solvay conference, in 1930, Einstein presented Bohr with another thought experiment, an imaginary contraption involving a spring scale with a box full of light hanging from it, timed with an accurate clock. Bohr, again, thought that Einstein was trying to get around a quantum uncertainty principle. After thinking for a while, Bohr revealed that Einstein's thought experiment "failed" because he (Einstein) had forgotten to take his own theory of general relativity into account.

This episode has become a legend in the history of quantum physics—Einstein hoisted by his own petard. But in reality, the problem was with Bohr. Einstein's thought experiment in 1930 was never intended to get around any kind of uncertainty principle—his complaint, once again, was locality, just as it had been at Solvay three years earlier. According to his friend Paul Ehrenfest, Einstein "absolutely no longer doubted the uncertainty relations" and had developed this thought experiment "for a totally different purpose." Bohr had once again missed the point.